# MetaCNVNet: A hybrid deep learning framework with multi-caller integration for enhanced copy number variation detection

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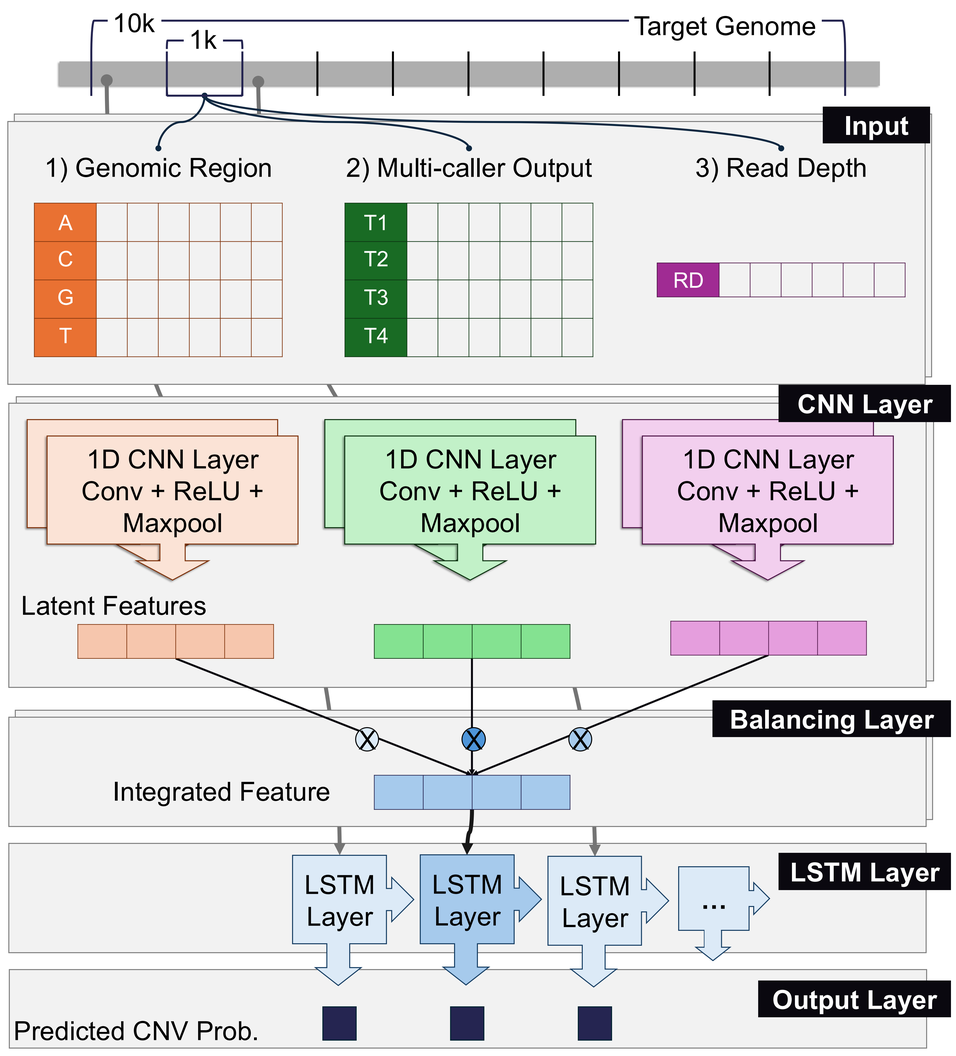
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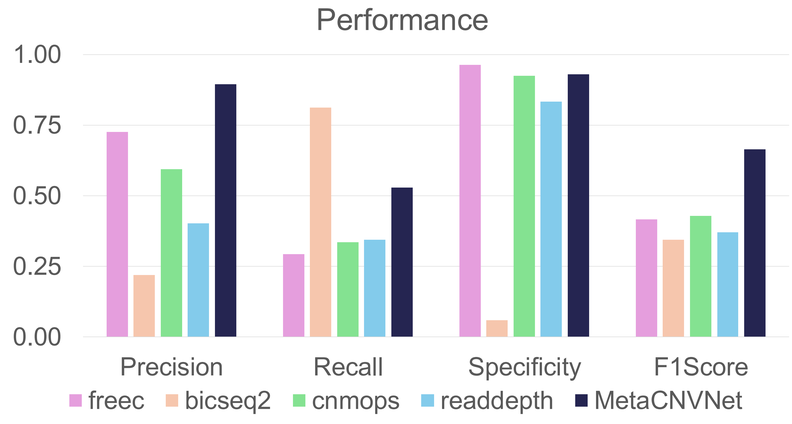
## Background

- **Copy number variations (CNVs):** Structural DNA alterations (kb–Mb) that drive genomic disorders and cancers via dosage effects and regulatory disruptions. CNVs are a major source of genomic diversity with strong clinical relevance.
- **Challenge:** Current CNV methods perform inconsistently, with variable sensitivity and specificity
- **Limitation:** No gold standard exists for reliable CNV detection across diverse settings, limiting consistent interpretation. Differences in sequencing platforms and sample quality further complicate detection.
- **Our Solution:** MetaCNVNet, a hybrid deep learning framework integrating multi-caller outputs with sequence and read-depth features, substantially improves detection precision, especially for complex CNVs.

## MetaCNVNet: CNN + LSTM Hybrid Model




- **Input**: Genomic sequences were one-hot encoded. Read depth (RD) signals computed from the read alignments, and multi-caller output aggregated from multiple CNV callers (READDEPTH[1], FREEC[2], CNVkit[3], CN.MOPS[4]).
- **CNN Layer**: Each input feature processed through two 1D conv layers with MaxPool to extract latent feature representations.
- **Balancing Layer**: Learns feature-specific weights and integrates them into a single latent vector for each 1kb bin.
- **LSTM Layer**: The weighted feature vectors were sequentially fed into an LSTM network to capture contextual dependencies across multiple adjacent 1kb bins (e.g., 10kb in total), enabling prediction over longer genomic regions.
- **Output Layer**: Final CNV probability predictions (0–1) were generated through a dense layer followed by a sigmoid activation.




## Conclusion

- MetaCNVNet outperforms CNV callers in precision, specificity, and F1-score
- Hybrid CNN–LSTM captures both local patterns and long-range dependencies and reduces false positives in CNV detection across diverse regions.
- Future extensions will target finer-resolution modeling, CNV subtype classification, and copy number estimation to broaden applicability.
- Integration with long-read sequencing and domain adaptation techniques could further expand MetaCNVNet's utility across diverse sample types and sequencing platforms.

## References

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[4] Klambauer G, Schwarzbauer K, Mayr A, Clevert DA, Mitterecker A, Bodenhofer U, Hochreiter S. cn.MOPS: mixture of Poissons for discovering copy number variations in next-generation sequencing data with a low false discovery rate. Nucleic Acids Res. 2012.

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